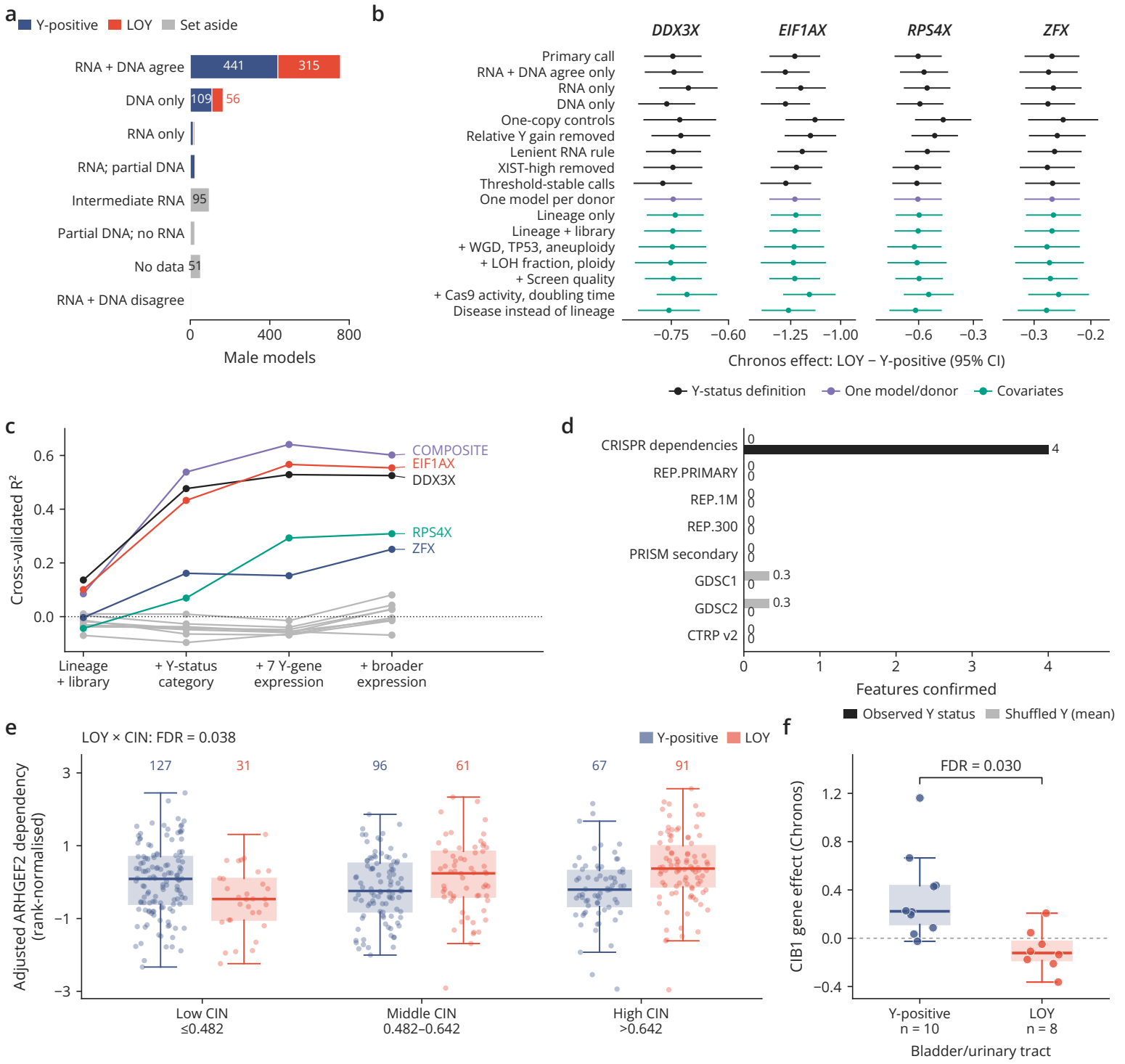

a
Y-positive
LOY
Set aside
RNA + DNA agree
441
315
DNA only
109
56
RNA only
RNA; partial DNA
Intermediate RNA
95
Partial DNA; no RNA
No data
51
RNA + DNA disagree
0
400
800
Male models
b
DDX3X
EIF1AX
RPS4X
ZFX
Primary call
RNA + DNA agree only
RNA only
DNA only
One-copy controls
Relative Y gain removed
Lenient RNA rule
XIST-high removed
Threshold-stable calls
One model per donor
Lineage only
Lineage + library
+ WGD, TP53, aneuploidy
+ LOH fraction, ploidy
+ Screen quality
+ Cas9 activity, doubling time
Disease instead of lineage
−0.75
−0.60
−1.25
−1.00
−0.6
−0.3
−0.3
−0.2
Chronos effect: LOY − Y-positive (95% CI)
Y-status definition
One model/donor
Covariates
c
Cross-validated R²
0.6
0.4
0.2
0.0
COMPOSITE
EIF1AX
DDX3X
RPS4X
ZFX
Lineage + library
+ Y-status category
+ 7 Y-gene expression
+ broader expression
d
CRISPR dependencies
0
4
REP.PRIMARY
0
0
REP.1M
0
0
REP.300
0
0
PRISM secondary
0
0
GDSC1
0.3
0
GDSC2
0.3
0
CTRP v2
0
0
0
1
2
3
4
Features confirmed
Observed Y status
Shuffled Y (mean)
e
LOY × CIN: FDR = 0.038
Y-positive
LOY
127
31
96
61
67
91
Adjusted ARHGEF2 dependency (rank-normalised)
3
1
−1
−3
Low CIN
≤0.482
Middle CIN
0.482–0.642
High CIN
>0.642
f
FDR = 0.030
CIB1 gene effect (Chronos)
1.2
0.8
0.4
0.0
−0.4
Y-positive
n = 10
LOY
n = 8
Bladder/urinary tract